

# Big Data Technologies - Detailed Learning Notes

## Phase 6: Big Data Fundamentals

Why we use it:

Traditional systems fail when data grows huge.

Topics:

- What is Big Data: huge datasets requiring distributed processing.
- 5Vs: Volume, Velocity, Variety, Veracity, Value.
- Structured vs Unstructured data.
- Batch vs Streaming processing.
- Distributed Systems Architecture:

Master node coordinates workers.

Workers process chunks in parallel.

Horizontal Scaling:

Add more machines.

Fault Tolerance:

Replication and recovery when nodes fail.

# Phase 7: Hadoop Fundamentals

Why:

Stores/processes massive distributed data.

Architecture:

Client → NameNode → DataNodes

Components:

- HDFS
- MapReduce
- YARN
- Common Utilities

Important:

NameNode: metadata manager

DataNode: stores blocks

Secondary NameNode: checkpointing

Replication: copies blocks

Rack Awareness: stores copies across racks

## Phase 8: HDFS

Why:

Distributed file storage.

Commands:

`hdfs dfs -ls`

`hdfs dfs -put`

`hdfs dfs -get`

`hdfs dfs -cat`

`hdfs dfs -rm`

Workflow:

Split → replicate → distribute → access.

# Phase 9: MapReduce

Why:

Parallel processing large datasets.

Architecture:

Input → Mapper → Shuffle/Sort → Reducer → Output

Concepts:

Mapper

Reducer

Combiner

Partitioner

InputSplit

Execution Flow

Examples:

Word Count

Log Analysis

Average Calculator

# Phase 10: Hive

Why:

SQL on Hadoop.

Architecture:

User → HiveQL → Compiler → Execution Engine → HDFS

Topics:

Managed vs External tables

Partitioning

Bucketing

Joins

UDFs

# Phase 11: HBase

Why:

NoSQL real-time database on Hadoop.

Architecture:

Client → HMaster → RegionServers → HDFS

Topics:

Column-family model

CRUD

Scans

Shell commands

# Phase 12: Spark + PySpark

Why:

Faster than MapReduce (in-memory processing).

Architecture:

Driver → Cluster Manager → Executors

Topics:

RDD

DataFrames

Transformations

Actions

Spark SQL

Optimization

Persistence

# Phase 13: Kafka

Why:

Real-time event streaming.

Architecture:

Producer → Broker → Topic Partition → Consumer

Topics:

Offsets

Partitions

Consumer groups

Streaming pipelines



## Phase 14: Airflow

Why:

Workflow scheduling.

Architecture:

Scheduler → DAG → Workers → Metadata DB

Topics:

Tasks

Operators

Scheduling

ETL automation

# Phase 15: Spark MLlib

Why:

Distributed machine learning.

Topics:

Preprocessing

Feature Engineering

Classification

Regression

Evaluation